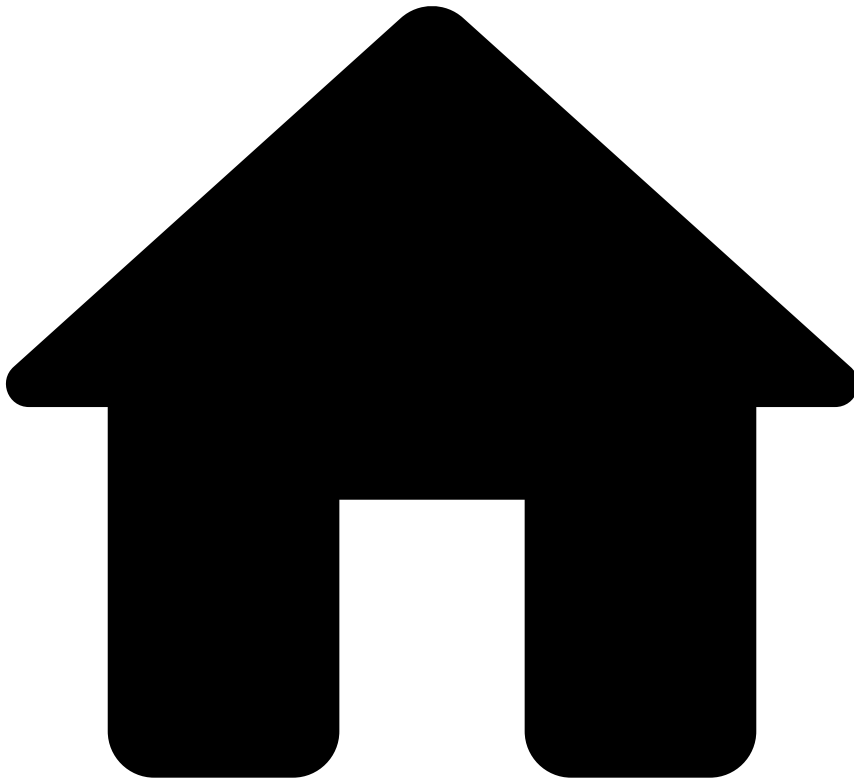


-



- [Out-of-the-Box Resources](#)
- Rules Projects
- Fraud Rules

On this page

Fraud Rules

Description🔍📄

Contains rules that look for standard fraud patterns. These rules are used to approve/decline a payment and optionally mark the event as an alert which can then be further investigated by fraud prevention agents in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

All-Payments-Velocity-FR1

Comment

Decline payment if the number of transactions per Customer ID and MID, in the last day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR1]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_count_1d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_count_1d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_count_1d"].threshold
?? 0;
```

Condition

```
// Limits
COUNT_LIMIT > 0
and
(event.cust_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}) in the last day ({event.cust_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR2

Comment

Decline payment if the number of transactions per Customer ID and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR2]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_2_max_customer_id_merchant_id_count_7d"].threshold
??
```

```
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr2_max_customer_id_merchant_id_count_7d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr2_max_customer_id_merchant_id_count_7d"].threshold
?? 0;
```

Condition

```
COUNT_LIMIT > 0
and
(event.cust_merch_cnt_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}) in the last week ({event.cust_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR3

Comment

Decline payment if the number of transactions per Customer Email and MID, in the past day, is above a set threshold

Actions

- [All-Payments-Velocity-FR3]

Variables

```
COUNT_LIMIT =
tenanted_rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_count_1d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_count_1d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_count_1d"].threshold
?? 0;
```

Condition

```
COUNT_LIMIT > 0
and
(event.email_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer Email ({event.customer_email}) and MID ({event.merchant_id}) in the last day ({event.email_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR4

Comment

Decline payment if the number of transactions per Customer Email and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR4]

Variables

```
COUNT_LIMIT =  
  
tenanted_rules_thresholds["all_payments_velocity_fr4_max_customer_email_merchant_id_count_7d"].threshold  
?? merchant_risk_levels__rules_thresholds  
["all_payments_velocity_fr4_max_customer_email_merchant_id_count_7d"].threshold  
??  
global_default__rules_thresholds["all_payments_velocity_fr4_max_customer_email_merchant_id_count_7d"].threshold  
?? 0;
```

Condition

```
COUNT_LIMIT > 0  
and  
(event.email_merch_cnt_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer Email ({event.customer_email}) and MID ({event.merchant_id}) in the last week ({event.email_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR5

Comment

Decline payment if the number of transactions per Device IP and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR5]

Variables

```
COUNT_LIMIT =  
tenanted_rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].threshold  
??  
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].threshold  
??  
global_default__rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].threshold
```

```
old
?? 0;
```

Condition

```
COUNT_LIMIT > 0
and
(event.devip_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same device ip ({event.device_ip}) and MID ({event.merchant_id}), in the last day ({event.devip_merch_cnt_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR7

Comment

Decline payments if the number of transactions per Shipping Address and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR7]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merchant_id_count_1d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merchant_id_count_1d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merchant_id_count_1d"].threshold
?? 0;
```

Condition

```
COUNT_LIMIT > 0
and
(event.shipaddr_merch_cnt_1d ?? 0) >= COUNT_LIMIT
// Full Shipping Address - Normalized and concatenated fields: shipping_addr_line1,
shipping_addr_line2, shipping_zip, shipping_city, shipping_region, shipping_country_code
```

Explanation

Number of transactions per Shipping Address ({event.full_shipping_address}) and MID ({event.merchant_id}) in the last day ({event.shipaddr_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR8

Comment

Decline payments if the number of transactions per Shipping Address and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR8]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merchant_id_count_7d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merchant_id_count_7d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merchant_id_count_7d"].threshold
?? 0;
```

Condition

```
COUNT_LIMIT > 0
and
(event.shipaddr_merch_cnt_7d ?? 0) >= COUNT_LIMIT
// Full Shipping Address - Normalized and concatenated fields: shipping_addr_line1,
shipping_addr_line2, shipping_zip, shipping_city, shipping_region, shipping_country_code
```

Explanation

Number of transactions for the same Shipping Address ({event.full_shipping_address}) and MID ({event.merchant_id}) in the last week ({event.shipaddr_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR9

Comment

Decline payment if the amount of transactions for the same Customer ID and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR9]

Variables

```
AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_sum_amount_unified_currency_1d"].threshold
??
```

```

merchant_risk_levels__rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_sum_amount_unified_currency_1d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_sum_amount_unified_currency_1d"].threshold
?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.cust_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;

```

Condition

```

AMOUNT_LIMIT > 0
and
(event.cust_merch_sum_amt_1d ?? 0) >= AMOUNT_LIMIT

```

Explanation

Amount of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}), in the last day ({localvar.AMOUNT_CONVERTED} {event.currency_code}), is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR10

Comment

Decline payment if the amount of transactions for the same Customer ID and MID, in the past week, is above a set threshold

Actions

- [All-Payments-Velocity-FR10]

Variables

```

AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_sum_amount_unified_currency_7d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_sum_amount_unified_currency_7d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_sum_amount_unified_currency_7d"].threshold
?? 0;

//for explanation
AMOUNT_CONVERTED = ((event.cust_merch_sum_amt_7d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0) /100;

```

Condition

```

AMOUNT_LIMIT > 0
and
event.cust_merch_sum_amt_7d ?? 0 >= AMOUNT_LIMIT

```

Explanation

Amount of transactions for the same Customer ID and MID ({event.merchant_id}) in the last week ({localvar.AMOUNT_CONVERTED}) is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR11

Comment

Decline payment if the amount of transactions for the same Customer Email and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR11]

Variables

```
AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.email_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;
```

Condition

```
AMOUNT_LIMIT > 0
and
(event.email_merch_sum_amt_1d ?? 0) >= AMOUNT_LIMIT
```

Explanation

Amount of transactions for the same Customer Email and MID ({event.merchant_id}) in the last day ({localvar.AMOUNT_CONVERTED} {event.currency_code}) is equal or above limit ({localvar.AMOUNT_LIMIT} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR12

Comment

Decline payment if the amount of transactions for the same Customer Email and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR12]

Variables

```
AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
??
global_default__rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.email_merch_sum_amt_7d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;
```

Condition

```
AMOUNT_LIMIT > 0
and
(event.email_merch_sum_amt_7d ?? 0) >= AMOUNT_LIMIT
```

Explanation

Amount of transactions for the same Customer Email and MID ({event.merchant_id}) in the last week ({localvar.AMOUNT_CONVERTED} {event.currency_code}) is equal or above limit ({localvar.AMOUNT_LIMIT} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR13

Comment

Decline payment if the card has a number of failed transactions, in the last 24 hours, above a set threshold.

Actions

- [Cards-Velocity-FR13]

Variables

```
COUNT_LIMIT =
tenanted_rules_thresholds["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
?? merchant_risk_levels__rules_thresholds
["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
??
global_default__rules_thresholds["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_card_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Card ({event.payment_card_id}) failed transactions ({event.fail_card_cnt_1d}), in the last day, are equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR14

Comment

Decline payment if the number of distinct cards with failed payments per MID, in the last minute, is above a set threshold

Actions

- [Cards-Velocity-FR14]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ??
  merchant_risk_levels__rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ??
  global_default__rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_dist_card_merch_1min ?? 0) >= COUNT_LIMIT
```

Explanation

Distinct failed Card transactions per MID ({event.fail_dist_card_merch_1min}), in the last minute, is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR15

Comment

Decline payment if the card has a number of transactions, in the last day, above a set threshold.

Actions

- [Cards-Velocity-FR15]

Variables

```
COUNT_LIMIT =
tenanted_rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_count_1d"].threshold
??
merchant_risk_levels__rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_count_1d"].threshold
??
global_default__rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_count_1d"].threshold
?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.card_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions ({event.card_merch_cnt_1d}) for the same Card ID ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day, is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR16

Comment

Decline payment if the amount of transactions per Card ID and MID, in the past day, is above a set threshold.

Actions

- [Cards-Velocity-FR16]

Variables

```
AMOUNT_LIMIT =
tenanted_rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
??
```

```

merchant_risk_levels__rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
?? global_default__rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
?? 0;

//for explanation
AMOUNT_CONVERTED = ((event.card_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0) /100;

```

Condition

```

// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
AMOUNT_LIMIT > 0
and
(event.card_merch_sum_amt_1d?? 0) >= AMOUNT_LIMIT

```

Explanation

Total amount transactions ({event.card_merch_cnt_1d}) for the same Card ID ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day, is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR17

Comment

Alert payment and flag card to be limited if multiple approved transactions with the same amount, in the same MID, in the last day. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Behavior-FR17]
- [Limit/Review_Card]
- Alert

Variables

```

COUNT_LIMIT =

tenanted_rules_thresholds["cards_behavior_fr17_max_appr_auth_payment_card_id_merchant_id_amount_
as_str_count_1d"].threshold
??
merchant_risk_levels__rules_thresholds["cards_behavior_fr17_max_appr_auth_payment_card_id_mercha
nt_id_amount_as_str_count_1d"].threshold
??
global_default__rules_thresholds["cards_behavior_1_max_appr_auth_payment_card_id_merchant_id_amo
unt_as_str_count_1d"].threshold
?? 0;
AMOUNT_LIMIT =

global_default__rules_thresholds["cards_behavior_fr17_max_amount_unified_currency"].threshold ??
0;
//for explanation
AMOUNT_CONVERTED = ((event.amount_unified_currency ?? 0) * 1.0) /100;

```

Condition

```
// Filters
event.payment_method_type == 'card'
and (event.payment_card_id ?? '' != '')
and COUNT_LIMIT > 0
and AMOUNT_LIMIT > 0
// Limits
and (event.appr_card_merch_amtstr_cnt_1d ?? 0) >= COUNT_LIMIT
and (event.amount_unified_currency ?? 0) >= AMOUNT_LIMIT // current transaction amount >= 20
```

Explanation

Card transactions with the same amount of {localvar.AMOUNT_CONVERTED} {event.currency_code}, in the same merchant ({event.appr_card_merch_amtstr_cnt_1d}), in the last 24h, are equal or above limit ({localvar.COUNT_LIMIT}). Card ID flagged to be limited.

Mark As

N/A

Cards-Behavior-FR18

Comment

Decline payment and block Customer Email if the number of distinct rejected cards per Customer Email, in the last 7 days, is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Behavior-FR18]
- [Auto_Decline_Email]

Variables

```
COUNT_LIMIT =
global_default__rules_thresholds["cards_behavior_fr18_max_failed_customer_email_countdist_payment_card_id_7d"].threshold ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_email_dist_card_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct cards ({event.fail_email_dist_card_7d}) with failed transaction per Customer Email, in the last week, is equal or above limit ({localvar.COUNT_LIMIT}). Customer Email ({event.customer_email}) is flagged to be auto declined

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR19

Comment

Decline and block Device IP if the number of distinct rejected cards per Device IP, in the last day, is above a set threshold. Depends on 'List Management Service' to put the IP in the configured list.

Actions

- [Auto_Decline_IP]
- [Cards-Behavior-FR19]

Variables

```
COUNT_LIMIT =  
global_default__rules_thresholds["cards_behavior_fr19_max_failed_ip_countdist_payment_card_id_1d"  
].threshold ?? 0;
```

Condition

```
// Filters  
event.payment_method_type == 'card'  
and  
(event.payment_card_id ?? '' != '')  
and  
// Limits  
COUNT_LIMIT > 0  
and  
(event.fail_ip_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Cards with failed transactions per Device IP ({event.device_ip}), in the last day ({event.fail_ip_dist_card_1d}), are equal or above limit {localvar.COUNT_LIMIT}. Device IP ({event.device_ip}) is flagged to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR20

Comment

Decline payment if the number of distinct emails for the same card and MID, in the last day, is above a set threshold.

Actions

- [Cards-Behavior-FR20]

Variables

```
COUNT_LIMIT =  
  
tenanted_rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_countdist_customer_email_1d"].threshold  
??  
merchant_risk_levels__rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_countdist_customer_email_1d"].threshold  
??
```

```
global_default__rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_countdist_customer_email_1d"].threshold
?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.card_merch_dist_email_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Customer Email per card ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day ({event.card_merch_dist_email_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR21

Comment

Decline payment if the number of distinct failed card transactions per Customer Email and MID, in the last week, is above a set threshold

Actions

- [Cards-Behavior-FR21]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
??
merchant_risk_levels__rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
??
global_default__rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_email_merch_dist_card_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of cards with failed transactions per Email ({event.customer_email}) and MID ({event.merchant_id}), in the last 24h ({event.fail_email_merch_dist_card_7d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR22

Comment

Alert payment if the number of distinct cards for the same Device ID and MID, in the last day, is above a set threshold.

Actions

- [Limit/Review_Device]
- Alert
- [Cards-Behavior-FR22]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold
  ??
  merchant_risk_levels__rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold
  ?? global_default__rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.device_id ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.merch_device_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Cards per Device ID ({event.device_id}) and MID ({event.merchant_id}), in the last day ({event.merch_device_dist_card_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-Behavior-FR23

Comment

Alert payment if the number of distinct cards for the same Billing Email and MID, in the last day, is above a set threshold.

Actions

- Alert
- [Cards-Behavior-FR23]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].threshold
??
merchant_risk_levels__rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].threshold
??
global_default__rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].threshold
?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.billing_email ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.merch_bill_email_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Cards per Billing Email ({event.billing_email}) and MID ({event.merchant_id}), in the last day ({event.merch_bill_email_dist_card_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-MCC-FR24

Comment

Decline payment if the number of small amount card transactions per MID with listed MCCs, in the last 15min, is above a set threshold.

Actions

- [Cards-MCC-FR24]

Variables

```
COUNT_LIMIT
=global_default__rules_thresholds["cards_behavior_fr24_merch_small_amt_cnt_15min"].threshold ??
0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.merchant_id ?? '' != '') and (event.merchant_mcc ?? '' != '')
and
// Limits
```

```
COUNT_LIMIT > 0
and (event.merch_small_amt_cnt_15min ?? 0) >= COUNT_LIMIT // trx amount <= 10
and card_testing_mcc.contains(event.merchant_mcc) // e.g Digital Goods, TaxiCabs, Streaming
Services
```

Explanation

MCC is listed ({event.merchant_mcc}) and multiple small amount ({event.amount_unified_currency}) card payments per MID ({event.merchant_id}), in the last 15m ({event.merch_small_amt_cnt_15min}), are equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-MCC-FR25

Comment

Decline payment if MCC is restricted for this BIN

Actions

- [Cards-MCC-FR25]

Variables

N/A

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_bin ?? '' != '') and (event.merchant_mcc ?? '' != '')
and
// Limits
restricted_mcc_per_bin.contains(event.merchant_mcc)
```

Explanation

Card payment with MCC ({event.merchant_mcc}) that is restricted for this BIN ({event.payment_card_bin})

Mark As

The outcome of this rule is: **decline**

Cards-MCC-FR26

Comment

Decline payment if the transaction amount is above MCC limit.

Actions

- [Cards-MCC-FR26]

Variables

```
MCC_AMOUNT_LIMIT = mcc_limits[event.merchant_mcc].amt_limit ?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.amount_unified_currency ?? 0) * 1.0) /100;
MCC_AMOUNT_LIMIT_CONVERTED = ((MCC_AMOUNT_LIMIT) * 1.0) / 100;
```

Condition

```
MCC_AMOUNT_LIMIT > 0
and
(event.amount_unified_currency ?? 0) >= MCC_AMOUNT_LIMIT
```

Explanation

Transaction amount ({localvar.AMOUNT_CONVERTED}) is equal or above this MCC limit ({localvar.MCC_AMOUNT_LIMIT_CONVERTED}) (MCC {event.merchant_mcc}).

Mark As

The outcome of this rule is: **decline**

Cards-CNP-FR27

Comment

Alert payment and flag a card to be limited if it has made multiple CNP transactions in a high number of unique merchants, in the last 5 minutes. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Limit/Review_Card]
- Alert
- [Cards-CNP-FR27]

Variables

```
COUNT_LIMIT =
global_default__rules_thresholds["cards_cnp_fr28_max_auth_cnp_payment_card_id_countdist_merchant_id_5m"].threshold ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
// Limits
(event.cnp_card_dist_merch_5min ?? 0) >= COUNT_LIMIT
```

Explanation

Card CNP transactions in unique merchants, in the last 5 minutes ({event.cnp_card_dist_merch_5min}), are equal or over the limit ({localvar.COUNT_LIMIT}). Card ({event.payment_card_id}) flagged to be limited.

Mark As

N/A

Cards-Mismatch-FR28

Comment

Alert a card payment for listed MIDs if card country mismatch merchant country

Actions

- [Cards-Mismatch-FR28]
- Alert

Variables

N/A

Condition

```
// Filters
event.payment_method_type == 'card'
and (event.payment_card_country_code ?? '' != '') and (event.merchant_country_code ?? '' != '')
and (event.merchant_id ?? '' != '')
//Limits
and listed_merchants__international_payments.contains(event.merchant_id)
and event.payment_card_country_code != event.merchant_country_code
```

Explanation

Mismatch between Card Country ({event.payment_card_country_code}) and Merchant Country ({event.merchant_country_code}) and MID ({event.merchant_id}) is limited to international card payments

Mark As

N/A

Cards-BinAttack-FR29

Comment

Ratio of distinct card_id and distinct expiry dates per BIN and Merchant in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-BinAttack-FR29]

Variables

```
// curve7 (th7) is expected to have better results then the others (good compromise between
number of alerts and detection rate)
binAttackThreshold = 7;

// merch_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 7 then return event.merch_cardbin_cur7;
end
end
```

Condition

```
thresholdDecision  
and !merchant_id_bin_attacks.contains(event.merchant_id ?? '') // Whitelisted Merchants to be  
excluded from this rule
```

Explanation

The ratio between distinct card_id ({event.merch_cardbin_cntdist_card_10m}) and distinct exp_date ({event.merch_cardbin_cntdist_cardexpd}), per BIN/Merchant on the last 10m, is below limit { {localvar.binAttackThreshold} }

Mark As

The outcome of this rule is: **decline**